Answer Key - Physics - Undergraduate Level Assessment 6 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 4; B) 5; C) 6; D) 20
   Note: Correct option: A - 4

2. **Answer:** D
   Options: A) Decreases by a factor of 81.; B) Decreases by a factor of 9.; C) Increases by a factor of 9.; D) Increases by a factor of 81.
   Note: Correct option: D - Increases by a factor of 81.

3. **Answer:** D
   Options: A) the weak nuclear force; B) the strong nuclear force; C) vacuum polarization; D) interactions with the ionic lattice
   Note: Correct option: D - interactions with the ionic lattice

4. **Answer:** D
   Options: A) 0.1 GeV/c^2; B) 0.2 GeV/c^2; C) 0.5 GeV/c^2; D) 1.0 GeV/c^2
   Note: Correct option: D - 1.0 GeV/c^2

5. **Answer:** A
   Options: A) Electron; B) Meson; C) Photon; D) Boson
   Note: Correct option: A - Electron

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '5'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '4'.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '250'.

Long Form Response

11. **Answer:** Electrons filling inner shell vacancies that are created in the metal atoms. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key